

CS 383

Chomsky Normal Form

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Chomsky Normal Form (CNF)

A CFG $G = (N, \Sigma, P, S)$ is in **Chomsky Normal Form** if all rules have one of these forms

- $S \rightarrow \varepsilon$ where S is the start variable
- $A \rightarrow BC$ where $A \in N$ and $B, C \in N \setminus \{S\}$
- $A \rightarrow t$ where $A \in N$ and $t \in \Sigma$

Note

- The only rule with ε on the right has the start variable on the left
- The start variable doesn't appear on the right hand side of any rule

CNF example

Let $A = \{w \mid w \in \{a, b\}^* \text{ and } w = w^R\}$.

CFG in CNF

$$S \rightarrow AU \mid BV \mid a \mid b \mid \varepsilon$$

$$T \rightarrow AU \mid BV \mid a \mid b$$

$$U \rightarrow TA \mid a$$

$$V \rightarrow TB \mid b$$

$$A \rightarrow a$$

$$B \rightarrow b$$

Derivation of baaab

S

CNF example

Let $A = \{w \mid w \in \{a, b\}^* \text{ and } w = w^R\}$.

CFG in CNF

$$S \rightarrow AU \mid BV \mid a \mid b \mid \varepsilon$$

$$T \rightarrow AU \mid BV \mid a \mid b$$

$$U \rightarrow TA \mid a$$

$$V \rightarrow TB \mid b$$

$$A \rightarrow a$$

$$B \rightarrow b$$

Derivation of baaab

$$S \Rightarrow BV$$

CNF example

Let $A = \{w \mid w \in \{a, b\}^* \text{ and } w = w^R\}$.

CFG in CNF

$$S \rightarrow AU \mid BV \mid a \mid b \mid \varepsilon$$

$$T \rightarrow AU \mid BV \mid a \mid b$$

$$U \rightarrow TA \mid a$$

$$V \rightarrow TB \mid b$$

$$A \rightarrow a$$

$$B \rightarrow b$$

Derivation of baaab

$$S \Rightarrow BV$$

$$\Rightarrow bV$$

CNF example

Let $A = \{w \mid w \in \{a, b\}^* \text{ and } w = w^R\}$.

CFG in CNF

$$S \rightarrow AU \mid BV \mid a \mid b \mid \varepsilon$$

$$T \rightarrow AU \mid BV \mid a \mid b$$

$$U \rightarrow TA \mid a$$

$$V \rightarrow TB \mid b$$

$$A \rightarrow a$$

$$B \rightarrow b$$

Derivation of baaab

$$S \Rightarrow BV$$

$$\Rightarrow bV$$

$$\Rightarrow bTB$$

CNF example

Let $A = \{w \mid w \in \{a, b\}^* \text{ and } w = w^R\}$.

CFG in CNF

$$S \rightarrow AU \mid BV \mid a \mid b \mid \varepsilon$$

$$T \rightarrow AU \mid BV \mid a \mid b$$

$$U \rightarrow TA \mid a$$

$$V \rightarrow TB \mid b$$

$$A \rightarrow a$$

$$B \rightarrow b$$

Derivation of baaab

$$S \Rightarrow BV$$

$$\Rightarrow bV$$

$$\Rightarrow bTB$$

$$\Rightarrow bAUB$$

CNF example

Let $A = \{w \mid w \in \{a, b\}^* \text{ and } w = w^R\}$.

CFG in CNF

$$S \rightarrow AU \mid BV \mid a \mid b \mid \varepsilon$$

$$T \rightarrow AU \mid BV \mid a \mid b$$

$$U \rightarrow TA \mid a$$

$$V \rightarrow TB \mid b$$

$$A \rightarrow a$$

$$B \rightarrow b$$

Derivation of baaab

$$S \Rightarrow BV$$

$$\Rightarrow bV$$

$$\Rightarrow bTB$$

$$\Rightarrow bAUB$$

$$\Rightarrow baUB$$

CNF example

Let $A = \{w \mid w \in \{a, b\}^* \text{ and } w = w^R\}$.

CFG in CNF

$$S \rightarrow AU \mid BV \mid a \mid b \mid \varepsilon$$

$$T \rightarrow AU \mid BV \mid a \mid b$$

$$U \rightarrow TA \mid a$$

$$V \rightarrow TB \mid b$$

$$A \rightarrow a$$

$$B \rightarrow b$$

Derivation of baaab

$$S \Rightarrow BV$$

$$\Rightarrow bV$$

$$\Rightarrow bTB$$

$$\Rightarrow bAUB$$

$$\Rightarrow baUB$$

$$\Rightarrow baTAB$$

CNF example

Let $A = \{w \mid w \in \{a, b\}^* \text{ and } w = w^R\}$.

CFG in CNF

$$S \rightarrow AU \mid BV \mid a \mid b \mid \varepsilon$$

$$T \rightarrow AU \mid BV \mid a \mid b$$

$$U \rightarrow TA \mid a$$

$$V \rightarrow TB \mid b$$

$$A \rightarrow a$$

$$B \rightarrow b$$

Derivation of baaab

$$S \Rightarrow BV$$

$$\Rightarrow bV$$

$$\Rightarrow bTB$$

$$\Rightarrow bAUB$$

$$\Rightarrow baUB$$

$$\Rightarrow baTAB$$

$$\Rightarrow baaAB$$

CNF example

Let $A = \{w \mid w \in \{a, b\}^* \text{ and } w = w^R\}$.

CFG in CNF

$$S \rightarrow AU \mid BV \mid a \mid b \mid \varepsilon$$

$$T \rightarrow AU \mid BV \mid a \mid b$$

$$U \rightarrow TA \mid a$$

$$V \rightarrow TB \mid b$$

$$A \rightarrow a$$

$$B \rightarrow b$$

Derivation of baaab

$$S \Rightarrow BV$$

$$\Rightarrow bV$$

$$\Rightarrow bTB$$

$$\Rightarrow bAUB$$

$$\Rightarrow baUB$$

$$\Rightarrow baTAB$$

$$\Rightarrow baaAB$$

$$\Rightarrow baaaB$$

CNF example

Let $A = \{w \mid w \in \{a, b\}^* \text{ and } w = w^R\}$.

CFG in CNF

$$S \rightarrow AU \mid BV \mid a \mid b \mid \varepsilon$$

$$T \rightarrow AU \mid BV \mid a \mid b$$

$$U \rightarrow TA \mid a$$

$$V \rightarrow TB \mid b$$

$$A \rightarrow a$$

$$B \rightarrow b$$

Derivation of baaab

$$S \Rightarrow BV$$

$$\Rightarrow bV$$

$$\Rightarrow bTB$$

$$\Rightarrow bAUB$$

$$\Rightarrow baUB$$

$$\Rightarrow baTAB$$

$$\Rightarrow baaAB$$

$$\Rightarrow baaaB$$

$$\Rightarrow baaab$$

Converting to CNF

Theorem

Every context-free language A is generated by some CFG in CNF.

Proof.

Given a CFG $G = (N, \Sigma, P, S)$ generating A , we construct a new CFG $G' = (N', \Sigma, P', S')$ in CNF generating A .

There are five steps.

START Add a new start variable

BIN Replace rules with RHS longer than two with multiple rules each of which has a RHS of length two

DEL- ϵ Remove all ϵ -rules ($A \rightarrow \epsilon$)

UNIT Remove all unit-rules ($A \rightarrow B$)

TERM Add a variable and rule for each terminal ($T \rightarrow t$) and replace terminals on the RHS of rules

Proof continued

In the following $x \in V \cup \Sigma$ and $u \in (\Sigma \cup V)^+$

START Add a new start variable S' and a rule $S' \rightarrow S$

Proof continued

In the following $x \in V \cup \Sigma$ and $u \in (\Sigma \cup V)^+$

START Add a new start variable S' and a rule $S' \rightarrow S$

BIN Replace each rule $A \rightarrow xu$ with the rules $A \rightarrow xA_1$ and $A_1 \rightarrow u$ and repeat until the RHS of every rule has length at most two

Proof continued

In the following $x \in V \cup \Sigma$ and $u \in (\Sigma \cup V)^+$

START Add a new start variable S' and a rule $S' \rightarrow S$

BIN Replace each rule $A \rightarrow xu$ with the rules $A \rightarrow xA_1$ and $A_1 \rightarrow u$ and repeat until the RHS of every rule has length at most two

DEL- ε For each rule of the form $A \rightarrow \varepsilon$ other than $S' \rightarrow \varepsilon$ remove $A \rightarrow \varepsilon$ and update all rules with A in the RHS

Proof continued

In the following $x \in V \cup \Sigma$ and $u \in (\Sigma \cup V)^+$

START Add a new start variable S' and a rule $S' \rightarrow S$

BIN Replace each rule $A \rightarrow xu$ with the rules $A \rightarrow xA_1$ and $A_1 \rightarrow u$ and repeat until the RHS of every rule has length at most two

DEL- ε For each rule of the form $A \rightarrow \varepsilon$ other than $S' \rightarrow \varepsilon$ remove $A \rightarrow \varepsilon$ and update all rules with A in the RHS

- $B \rightarrow A$. Add rule $B \rightarrow \varepsilon$ unless $B \rightarrow \varepsilon$ has already been removed

Proof continued

In the following $x \in V \cup \Sigma$ and $u \in (\Sigma \cup V)^+$

START Add a new start variable S' and a rule $S' \rightarrow S$

BIN Replace each rule $A \rightarrow xu$ with the rules $A \rightarrow xA_1$ and $A_1 \rightarrow u$ and repeat until the RHS of every rule has length at most two

DEL- ε For each rule of the form $A \rightarrow \varepsilon$ other than $S' \rightarrow \varepsilon$ remove $A \rightarrow \varepsilon$ and update all rules with A in the RHS

- $B \rightarrow A$. Add rule $B \rightarrow \varepsilon$ unless $B \rightarrow \varepsilon$ has already been removed
- $B \rightarrow AA$. Add rule $B \rightarrow A$ and if $B \rightarrow \varepsilon$ has not already been removed, add it

Proof continued

In the following $x \in V \cup \Sigma$ and $u \in (\Sigma \cup V)^+$

START Add a new start variable S' and a rule $S' \rightarrow S$

BIN Replace each rule $A \rightarrow xu$ with the rules $A \rightarrow xA_1$ and $A_1 \rightarrow u$ and repeat until the RHS of every rule has length at most two

DEL- ε For each rule of the form $A \rightarrow \varepsilon$ other than $S' \rightarrow \varepsilon$ remove $A \rightarrow \varepsilon$ and update all rules with A in the RHS

- $B \rightarrow A$. Add rule $B \rightarrow \varepsilon$ unless $B \rightarrow \varepsilon$ has already been removed
- $B \rightarrow AA$. Add rule $B \rightarrow A$ and if $B \rightarrow \varepsilon$ has not already been removed, add it
- $B \rightarrow xA$ or $B \rightarrow Ax$. Add rule $B \rightarrow x$

Proof continued

In the following $x \in V \cup \Sigma$ and $u \in (\Sigma \cup V)^+$

START Add a new start variable S' and a rule $S' \rightarrow S$

BIN Replace each rule $A \rightarrow xu$ with the rules $A \rightarrow xA_1$ and $A_1 \rightarrow u$ and repeat until the RHS of every rule has length at most two

DEL- ε For each rule of the form $A \rightarrow \varepsilon$ other than $S' \rightarrow \varepsilon$ remove $A \rightarrow \varepsilon$ and update all rules with A in the RHS

- $B \rightarrow A$. Add rule $B \rightarrow \varepsilon$ unless $B \rightarrow \varepsilon$ has already been removed
- $B \rightarrow AA$. Add rule $B \rightarrow A$ and if $B \rightarrow \varepsilon$ has not already been removed, add it
- $B \rightarrow xA$ or $B \rightarrow Ax$. Add rule $B \rightarrow x$

UNIT For each rule $A \rightarrow B$, remove it and add rules $A \rightarrow u$ for each $B \rightarrow u$ unless $A \rightarrow u$ is a unit rule already removed

Proof continued

In the following $x \in V \cup \Sigma$ and $u \in (\Sigma \cup V)^+$

START Add a new start variable S' and a rule $S' \rightarrow S$

BIN Replace each rule $A \rightarrow xu$ with the rules $A \rightarrow xA_1$ and $A_1 \rightarrow u$ and repeat until the RHS of every rule has length at most two

DEL- ε For each rule of the form $A \rightarrow \varepsilon$ other than $S' \rightarrow \varepsilon$ remove $A \rightarrow \varepsilon$ and update all rules with A in the RHS

- $B \rightarrow A$. Add rule $B \rightarrow \varepsilon$ unless $B \rightarrow \varepsilon$ has already been removed
- $B \rightarrow AA$. Add rule $B \rightarrow A$ and if $B \rightarrow \varepsilon$ has not already been removed, add it
- $B \rightarrow xA$ or $B \rightarrow Ax$. Add rule $B \rightarrow x$

UNIT For each rule $A \rightarrow B$, remove it and add rules $A \rightarrow u$ for each $B \rightarrow u$ unless $A \rightarrow u$ is a unit rule already removed

TERM For each $t \in \Sigma$, add a new variable T and a rule $T \rightarrow t$; replace each t in the RHS of nonunit rules with T

Proof continued

In the following $x \in V \cup \Sigma$ and $u \in (\Sigma \cup V)^+$

START Add a new start variable S' and a rule $S' \rightarrow S$

BIN Replace each rule $A \rightarrow xu$ with the rules $A \rightarrow xA_1$ and $A_1 \rightarrow u$ and repeat until the RHS of every rule has length at most two

DEL- ε For each rule of the form $A \rightarrow \varepsilon$ other than $S' \rightarrow \varepsilon$ remove $A \rightarrow \varepsilon$ and update all rules with A in the RHS

- $B \rightarrow A$. Add rule $B \rightarrow \varepsilon$ unless $B \rightarrow \varepsilon$ has already been removed
- $B \rightarrow AA$. Add rule $B \rightarrow A$ and if $B \rightarrow \varepsilon$ has not already been removed, add it
- $B \rightarrow xA$ or $B \rightarrow Ax$. Add rule $B \rightarrow x$

UNIT For each rule $A \rightarrow B$, remove it and add rules $A \rightarrow u$ for each $B \rightarrow u$ unless $A \rightarrow u$ is a unit rule already removed

TERM For each $t \in \Sigma$, add a new variable T and a rule $T \rightarrow t$; replace each t in the RHS of nonunit rules with T

Each of the five steps preserves the language generated by the grammar so $L(G') = A$.



Example

Convert to CNF

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

START:

Example

Convert to CNF

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

START:

$$S \rightarrow A$$

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

Example

Convert to CNF

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

START:

$$S \rightarrow A$$

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

BIN: Replace $A \rightarrow BAB$:

Example

Convert to CNF

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

START:

$$S \rightarrow A$$

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

BIN: Replace $A \rightarrow BAB$:

$$S \rightarrow A$$

$$A \rightarrow BA_1 \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

$$A_1 \rightarrow AB$$

Example

Convert to CNF

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

DEL- ε : Remove $A \rightarrow \varepsilon$:

START:

$$S \rightarrow A$$

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

BIN: Replace $A \rightarrow BAB$:

$$S \rightarrow A$$

$$A \rightarrow BA_1 \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

$$A_1 \rightarrow AB$$

Example

Convert to CNF

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

START:

$$S \rightarrow A$$

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

BIN: Replace $A \rightarrow BAB$:

$$S \rightarrow A$$

$$A \rightarrow BA_1 \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

$$A_1 \rightarrow AB$$

DEL- ε : Remove $A \rightarrow \varepsilon$:

$$S \rightarrow A \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B$$

$$B \rightarrow 00 \mid \varepsilon$$

$$A_1 \rightarrow AB \mid B$$

Example

Convert to CNF

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

START:

$$S \rightarrow A$$

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

BIN: Replace $A \rightarrow BAB$:

$$S \rightarrow A$$

$$A \rightarrow BA_1 \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

$$A_1 \rightarrow AB$$

DEL- ε : Remove $A \rightarrow \varepsilon$:

$$S \rightarrow A \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B$$

$$B \rightarrow 00 \mid \varepsilon$$

$$A_1 \rightarrow AB \mid B$$

Remove $B \rightarrow \varepsilon$:

Example

Convert to CNF

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

START:

$$S \rightarrow A$$

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

BIN: Replace $A \rightarrow BAB$:

$$S \rightarrow A$$

$$A \rightarrow BA_1 \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

$$A_1 \rightarrow AB$$

DEL- ε : Remove $A \rightarrow \varepsilon$:

$$S \rightarrow A \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B$$

$$B \rightarrow 00 \mid \varepsilon$$

$$A_1 \rightarrow AB \mid B$$

Remove $B \rightarrow \varepsilon$:

$$S \rightarrow A \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A \mid \varepsilon$$

Don't add $A \rightarrow \varepsilon$ because we already removed it

Example

Convert to CNF

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

START:

$$S \rightarrow A$$

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

BIN: Replace $A \rightarrow BAB$:

$$S \rightarrow A$$

$$A \rightarrow BA_1 \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

$$A_1 \rightarrow AB$$

DEL- ε : Remove $A \rightarrow \varepsilon$:

$$S \rightarrow A \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B$$

$$B \rightarrow 00 \mid \varepsilon$$

$$A_1 \rightarrow AB \mid B$$

Remove $A_1 \rightarrow \varepsilon$:

Remove $B \rightarrow \varepsilon$:

$$S \rightarrow A \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A \mid \varepsilon$$

Don't add $A \rightarrow \varepsilon$ because we already removed it

Example

Convert to CNF

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

START:

$$S \rightarrow A$$

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

BIN: Replace $A \rightarrow BAB$:

$$S \rightarrow A$$

$$A \rightarrow BA_1 \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

$$A_1 \rightarrow AB$$

DEL- ε : Remove $A \rightarrow \varepsilon$:

$$S \rightarrow A \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B$$

$$B \rightarrow 00 \mid \varepsilon$$

$$A_1 \rightarrow AB \mid B$$

Remove $B \rightarrow \varepsilon$:

$$S \rightarrow A \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A \mid \varepsilon$$

Don't add $A \rightarrow \varepsilon$ because we already removed it

Remove $A_1 \rightarrow \varepsilon$:

$$S \rightarrow A \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Don't add $A \rightarrow \varepsilon$ because we already removed it

Example

Convert to CNF

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

START:

$$S \rightarrow A$$

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

BIN: Replace $A \rightarrow BAB$:

$$S \rightarrow A$$

$$A \rightarrow BA_1 \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

$$A_1 \rightarrow AB$$

DEL- ε : Remove $A \rightarrow \varepsilon$:

$$S \rightarrow A \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B$$

$$B \rightarrow 00 \mid \varepsilon$$

$$A_1 \rightarrow AB \mid B$$

Remove $B \rightarrow \varepsilon$:

$$S \rightarrow A \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A \mid \varepsilon$$

Don't add $A \rightarrow \varepsilon$ because we already removed it

Remove $A_1 \rightarrow \varepsilon$:

$$S \rightarrow A \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Don't add $A \rightarrow \varepsilon$ because we already removed it

UNIT: Remove $S \rightarrow A$

Example

Convert to CNF

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

START:

$$S \rightarrow A$$

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

BIN: Replace $A \rightarrow BAB$:

$$S \rightarrow A$$

$$A \rightarrow BA_1 \mid B \mid \varepsilon$$

$$B \rightarrow 00 \mid \varepsilon$$

$$A_1 \rightarrow AB$$

DEL- ε : Remove $A \rightarrow \varepsilon$:

$$S \rightarrow A \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B$$

$$B \rightarrow 00 \mid \varepsilon$$

$$A_1 \rightarrow AB \mid B$$

Remove $B \rightarrow \varepsilon$:

$$S \rightarrow A \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A \mid \varepsilon$$

Don't add $A \rightarrow \varepsilon$ because we already removed it

Remove $A_1 \rightarrow \varepsilon$:

$$S \rightarrow A \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Don't add $A \rightarrow \varepsilon$ because we already removed it

UNIT: Remove $S \rightarrow A$

$$S \rightarrow BA_1 \mid B \mid A_1 \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Example continued

From previous slide

$$S \rightarrow BA_1 \mid B \mid A_1 \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Example continued

From previous slide

$$S \rightarrow BA_1 \mid B \mid A_1 \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Remove $S \rightarrow B$

Example continued

From previous slide

$$S \rightarrow BA_1 \mid B \mid A_1 \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Remove $S \rightarrow B$

$$S \rightarrow BA_1 \mid A_1 \mid \varepsilon \mid 00$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Example continued

From previous slide

$$S \rightarrow BA_1 \mid B \mid A_1 \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Remove $S \rightarrow A_1$

Remove $S \rightarrow B$

$$S \rightarrow BA_1 \mid A_1 \mid \varepsilon \mid 00$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Example continued

From previous slide

$$S \rightarrow BA_1 \mid B \mid A_1 \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Remove $S \rightarrow A_1$

$$S \rightarrow BA_1 \mid \varepsilon \mid 00 \mid AB$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Remove $S \rightarrow B$

$$S \rightarrow BA_1 \mid A_1 \mid \varepsilon \mid 00$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Don't add $S \rightarrow B$ or $S \rightarrow A$

because we removed them

Example continued

From previous slide

$$S \rightarrow BA_1 \mid B \mid A_1 \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Remove $S \rightarrow A_1$

$$S \rightarrow BA_1 \mid \varepsilon \mid 00 \mid AB$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Remove $S \rightarrow B$

$$S \rightarrow BA_1 \mid A_1 \mid \varepsilon \mid 00$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Don't add $S \rightarrow B$ or $S \rightarrow A$
because we removed them

Remove $A \rightarrow B$

Example continued

From previous slide

$$S \rightarrow BA_1 \mid B \mid A_1 \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Remove $S \rightarrow A_1$

$$S \rightarrow BA_1 \mid \varepsilon \mid 00 \mid AB$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Remove $S \rightarrow B$

$$S \rightarrow BA_1 \mid A_1 \mid \varepsilon \mid 00$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Don't add $S \rightarrow B$ or $S \rightarrow A$
because we removed them

Remove $A \rightarrow B$

$$S \rightarrow BA_1 \mid \varepsilon \mid 00 \mid AB$$

$$A \rightarrow BA_1 \mid A_1 \mid 00$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Example continued

From previous slide

$$S \rightarrow BA_1 \mid B \mid A_1 \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Remove $S \rightarrow A_1$

$$S \rightarrow BA_1 \mid \varepsilon \mid 00 \mid AB$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Remove $A \rightarrow A_1$

Remove $S \rightarrow B$

$$S \rightarrow BA_1 \mid A_1 \mid \varepsilon \mid 00$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Don't add $S \rightarrow B$ or $S \rightarrow A$
because we removed them

Remove $A \rightarrow B$

$$S \rightarrow BA_1 \mid \varepsilon \mid 00 \mid AB$$

$$A \rightarrow BA_1 \mid A_1 \mid 00$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Example continued

From previous slide

$$S \rightarrow BA_1 \mid B \mid A_1 \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Remove $S \rightarrow A_1$

$$S \rightarrow BA_1 \mid \varepsilon \mid 00 \mid AB$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Remove $A \rightarrow A_1$

$$S \rightarrow BA_1 \mid \varepsilon \mid 00 \mid AB$$

$$A \rightarrow BA_1 \mid 00 \mid AB$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Remove $S \rightarrow B$

$$S \rightarrow BA_1 \mid A_1 \mid \varepsilon \mid 00$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Don't add $S \rightarrow B$ or $S \rightarrow A$ because we removed them

Don't add $A \rightarrow B$ because we removed it

Don't add $A \rightarrow A$ because it's useless

Remove $A \rightarrow B$

$$S \rightarrow BA_1 \mid \varepsilon \mid 00 \mid AB$$

$$A \rightarrow BA_1 \mid A_1 \mid 00$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Example continued

From previous slide

$$S \rightarrow BA_1 \mid B \mid A_1 \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Remove $S \rightarrow A_1$

$$S \rightarrow BA_1 \mid \varepsilon \mid 00 \mid AB$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Remove $A \rightarrow A_1$

$$S \rightarrow BA_1 \mid \varepsilon \mid 00 \mid AB$$

$$A \rightarrow BA_1 \mid 00 \mid AB$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Remove $S \rightarrow B$

$$S \rightarrow BA_1 \mid A_1 \mid \varepsilon \mid 00$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

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Remove $A \rightarrow B$

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$$A \rightarrow BA_1 \mid A_1 \mid 00$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Don't add $A \rightarrow B$ because we removed it
Don't add $A \rightarrow A$ because it's useless

Remove $A_1 \rightarrow B$

Example continued

From previous slide

$$S \rightarrow BA_1 \mid B \mid A_1 \mid \varepsilon$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Remove $S \rightarrow A_1$

$$S \rightarrow BA_1 \mid \varepsilon \mid 00 \mid AB$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Remove $A \rightarrow A_1$

$$S \rightarrow BA_1 \mid \varepsilon \mid 00 \mid AB$$

$$A \rightarrow BA_1 \mid 00 \mid AB$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Remove $S \rightarrow B$

$$S \rightarrow BA_1 \mid A_1 \mid \varepsilon \mid 00$$

$$A \rightarrow BA_1 \mid B \mid A_1$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid B \mid A$$

Don't add $S \rightarrow B$ or $S \rightarrow A$ because we removed them

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$$S \rightarrow BA_1 \mid \varepsilon \mid 00 \mid AB$$

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Don't add $A \rightarrow B$ because we removed it
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Remove $A_1 \rightarrow B$

$$S \rightarrow BA_1 \mid \varepsilon \mid 00 \mid AB$$

$$A \rightarrow BA_1 \mid 00 \mid AB$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid A \mid 00$$

Example continued

Copied from the previous slide

$$S \rightarrow BA_1 \mid \varepsilon \mid 00 \mid AB$$

$$A \rightarrow BA_1 \mid 00 \mid AB$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid A \mid 00$$

Remove $A_1 \rightarrow A$

Example continued

Copied from the previous slide

$$S \rightarrow BA_1 \mid \varepsilon \mid 00 \mid AB$$

$$A \rightarrow BA_1 \mid 00 \mid AB$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid A \mid 00$$

Remove $A_1 \rightarrow A$

$$S \rightarrow BA_1 \mid \varepsilon \mid 00 \mid AB$$

$$A \rightarrow BA_1 \mid 00 \mid AB$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid 00 \mid BA_1$$

Example continued

Copied from the previous slide

$$S \rightarrow BA_1 \mid \varepsilon \mid 00 \mid AB$$

$$A \rightarrow BA_1 \mid 00 \mid AB$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid A \mid 00$$

Remove $A_1 \rightarrow A$

$$S \rightarrow BA_1 \mid \varepsilon \mid 00 \mid AB$$

$$A \rightarrow BA_1 \mid 00 \mid AB$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid 00 \mid BA_1$$

TERM: Add $Z \rightarrow 0$

Example continued

Copied from the previous slide

$$S \rightarrow BA_1 \mid \varepsilon \mid 00 \mid AB$$

$$A \rightarrow BA_1 \mid 00 \mid AB$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid A \mid 00$$

Remove $A_1 \rightarrow A$

$$S \rightarrow BA_1 \mid \varepsilon \mid 00 \mid AB$$

$$A \rightarrow BA_1 \mid 00 \mid AB$$

$$B \rightarrow 00$$

$$A_1 \rightarrow AB \mid 00 \mid BA_1$$

TERM: Add $Z \rightarrow 0$

$$S \rightarrow BA_1 \mid \varepsilon \mid ZZ \mid AB$$

$$A \rightarrow BA_1 \mid ZZ \mid AB$$

$$B \rightarrow ZZ$$

$$A_1 \rightarrow AB \mid ZZ \mid BA_1$$

$$Z \rightarrow 0$$

Caution

Kozen gives a different procedure

- ① No START step; *This step is omitted entirely!*
- ② DEL- ε
- ③ UNIT
- ④ TERM
- ⑤ BIN

This procedure works¹ but can lead to an exponential blow up in the number of rules!

In general, if DEL- ε comes before BIN, then $|G'|$ is $O(2^{2^{|G|}})$;
if BIN comes before DEL- ε , then $|G'|$ is $O(|G|^2)$

UNIT is responsible for the quadratic blow up

So use whichever procedure you'd like, but Kozen's can be *very* bad
(Kozen's is bad if you have long rules with lots of variables with ε -rules)

¹for languages that don't contain ε

Example blow up

$$A \rightarrow BCDEEDCB \mid CBEDDEBC$$

$$B \rightarrow 0 \mid \varepsilon$$

$$C \rightarrow 1 \mid \varepsilon$$

$$D \rightarrow 2 \mid \varepsilon$$

$$E \rightarrow 3 \mid \varepsilon$$

has five variables and 10 rules

Converting using START, BIN, DEL- ε , UNIT, TERM gives a CFG with 18 variables and 125 rules

Example blow up

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$$E \rightarrow 3 \mid \varepsilon$$

has five variables and 10 rules

Converting using START, BIN, DEL- ε , UNIT, TERM gives a CFG with 18 variables and 125 rules

Converting using START, DEL- ε , UNIT, TERM, BIN gives a CFG with 1394 variables and 1953 rules

Bigger consequence of Kozen's procedure

Kozen does not allow *any* rules of the form $A \rightarrow \varepsilon$

This has two consequences

- 1 Every rule has one of two forms

$$A \rightarrow BC \quad \text{for } A, B, C \in N$$

$$A \rightarrow t \quad \text{for } A \in N \text{ and } t \in \Sigma$$

Note that there are no restrictions on the RHS of the rules $A \rightarrow BC$ whereas the procedure I outlined doesn't allow B or C to be the start nonterminal S

- 2 No rule can produce ε . Kozen's procedure turns a CFG G into a CFG G' where $L(G') = L(G) \setminus \{\varepsilon\}$